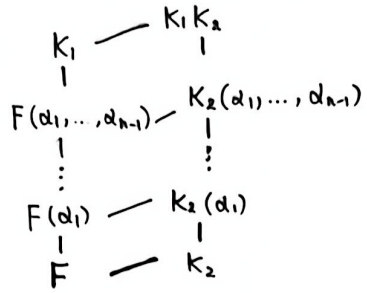
Lemma 1: $K/E/F$, $\alpha \in K$. $\Rightarrow m_{\alpha,E}(x) \mid m_{\alpha,F}(x)$, thus $[E(\alpha):E] \le [F(\alpha):F]$

Lemma 2: Note K_1K_2: minimal field that contains both K_1 and K_2.
$F \subseteq K_i \subseteq K$. and $[K_i:F] < \infty$, then $[K_1K_2:F] \le [K_1:F][K_2:F]$

pf:



$[K_2(\alpha_1):K_2] \le [F(\alpha_1):F]$

$[K_2(\alpha_1,\alpha_2):K_2(\alpha_1)] \le [F(\alpha_1,\alpha_2):F(\alpha_1)]$

Taking the product ■

Back to the proof: let $K = F(\alpha_1)F(\alpha_2)\dots F(\alpha_n) \Rightarrow [K:F] \le \prod_{k=1}^{n} [F(\alpha_k):F] < \infty$ □

Definition 3: K be finite extension ~~over~~ of F, $\{\alpha \in K \mid \alpha$ is algebraic over $F\}$ is a subfield of K called the algebraic closure of F in K.

abcde.

Thm: If L/K & K/F are algebraic, then L/F is algebraic.

Definition 4: K/F is ~~a~~ called a splitting field of $f(x)$ over F if.
(1) $f(x)$ splits completely in $K[x]$: $f(x) = (x-\alpha_1)\dots(x-\alpha_n)$ $\alpha_i \in K$, $\deg f = n$
(2) $K = F(\alpha_1,\dots,\alpha_n)$.

Thm: (1) For any field F and $f(x) \in F[x]$ of degree n, a splitting field K of F exists.
~~th~~ and $[K:F] \le n!$

(2) Let $\eta: F_1 \xrightarrow{\cong} F'$, $f(x) \in F[x]$, $f'(x) = \eta(f(x))$.
If E is a splitting field of $f(x)$ over F. E'...
then there exists an isomorphism $\sigma: E \xrightarrow{\cong} E'$ restricting to $\eta: F \xrightarrow{\cong} F'$

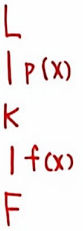
pf: (sketched)

Normal extension:
Def: An algebraic extension K of F is called normal if
for any irreducible polynomial $f(x) \in F[x]$ that has one zero in K, $f(x)$ splits completely over K.

Thm: A finite extension K of F is normal iff it's the splitting field of some $f(x) \in F[x]$

pf: $\Rightarrow$: $K = F(\alpha_1,\dots,\alpha_r)$ for $\alpha_1,\dots,\alpha_r \in K$, $\Rightarrow m_{\alpha_i}(x)$ over F splits completely
thus K is the splitting field of $m_{\alpha_1}(x)\dots m_{\alpha_r}(x)$ over F

$\Leftarrow$: Assume that K is a splitting field of $f(x) \in F[x]$ over F.
If $p(x) \in F[x]$ is an irreducible poly that has a zero $\alpha \in K$.
let L be the splitting field of $p(x)$ over K.
then L is the splitting field of $f(x)p(x)$ over ~~K~~ F.



We've introduced basic concepts and theorems of groups. rings especially homomorphism and isomorphism. then we'd like to consider fields.

Definition 1: 1) F is called a ~~ring~~ field if F is a ring, commutative, has multiplicative inverse for all $a \neq 0$.

2) F & G are fields, we call G over F if $F \subseteq G$, denoted by G/F.

3). $\begin{cases} \text{char}(F): \text{smallest prime } p, \text{ such that } \underbrace{1+1+\cdots+1}_{p} = 0. \\ \text{The prime field of } F: \text{smallest field of } F \text{ containing } 1_F: \begin{cases} \mathbb{F}_p & \text{char}(F) = p \\ \mathbb{Q} & \text{char}(F) = 0 \end{cases} \end{cases}$

Remark. 1. Since $F \subseteq G$, one can view G as the vector space over F. thus we can define $[G:F] = \dim_F G$.

And it's easy to check that $G/F/E \Rightarrow [G:E] = [G:F][F:E]$

2. A homomorphism $\phi: F \to E$ must be injective

Construction of field extensions.

let F be a field. $p(x)$ be an irreducible poly over F. with $\deg p(x) = n$

thus $K := F[x]/(p(x))$ is a field. take $\theta := x \pmod{(p(x))}$. then $K = \text{span}\{\theta, \theta^2, \cdots, \theta^{n-1}, 1\}$

Remark. 1. $p(x)$ over K has a zero, i.e. $p(\theta) = 0$ over K

2. Some examples: ① $\mathbb{R}[x]/(x^2+1) \longrightarrow \mathbb{C}$
$a + bx \longmapsto a + bi$

② $\mathbb{F}_2[x]/(x^2+x+1) \longrightarrow \mathbb{R}$, where $\mathbb{F}_2[x]/(x^2+x+1) = \{0, 1, x, x+1\} \cong \mathbb{F}_4$

Definition 2: let K/F. $\alpha_1, \cdots, \alpha_n \in K$.

(1) If $K = F(\alpha)$ for some $\alpha \in K$. we call K is a simple extension of F

(2) If $K = F(\alpha_1, \cdots \alpha_n)$ we call K a finitely generated extension of F.

Theorem 1: K/F, $\alpha \in K$. then

(1) either $1, \alpha, \alpha^2, \cdots$ are linear independent over F. then $F(\alpha) \cong \text{Frac}(F[x])$

(2) or $1, \alpha, \alpha^2, \cdots$, are linear dependent over F. then there exists the minimal poly of α over F. s.t. $m_\alpha(\alpha) = 0$. then $F(\alpha) = F[x]/(m_\alpha(x))$

pf: consider the homomorphism $\phi: F[x] \longrightarrow K$ $\quad f(x) \longmapsto f(\alpha)$

then in (1) we have $\text{Im}\,\phi = F(\alpha) \cong \text{Frac}(f[x])$

in (2) we have $\text{Im}\,\phi = F(\alpha) \cong F[x]/\ker\phi = F[x]/(m_\alpha(x))$ □.

Thm 2: TFAE (1) K is a finite extension of F; (2) K is finitely generated & algebraic over F

pf: (1) $\Rightarrow$ (2) is trivial.

for the uniqueness of splitting fields.

ι F, $E \not\cong F'$ and we have $\eta: F \xrightarrow{\cong} F'$

then $\exists \sigma: F(\alpha) \to E'$ injective restricting to η

iff $\eta(m_\alpha(x))$ has a root on E'

pf. $\Rightarrow$: let $\sigma(\alpha) = \beta$. then $\eta(m_\alpha(\beta)) = \eta(m_\alpha(\sigma(\alpha))) = \sigma(m_\alpha(\alpha)) = 0$

$\Leftarrow$: if $\eta(m_\alpha(\beta)) = 0$, $\beta \in E'$. then $F(\alpha) \cong F[x]/(f(x)) \xrightarrow{\eta} F'[x]/(\eta f(x)) \cong F'(\beta)$

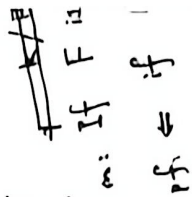
$\quad\quad\quad

Definition: If K is an algebraic extension K of F, a normal closure of

a field extension L of K such that

(1) L is a normal extension of F

(2) L is minimal, i.e if $L \supseteq L' \supseteq K$ is so that L'/F is normal then $L = L'$



Lemma: A normal closure of a finite extension K over F exists and is unique up to some isomorphisms

pf: (Existence): Assume $K = F(\alpha_1, \dots, \alpha_r)$. Put $f(x) = \prod_{i=1}^{r} m_{\alpha_i, F}(x) \in F[x]$

Take L to be a splitting field of $f(x)$ over K.

then L is a normal closure of K over F $\quad (L/K/F)$

(Uniqueness): L' is another normal closure of K over F

$$\begin{array}{ccc} L & \hookrightarrow & L' \\ \cup & & \cup \\ F & = & F \end{array}$$

by Thm: there exists an embedding $L \hookrightarrow L' \Rightarrow L = L'$ ▨

Perfection fields:

Define: 1) Frobenius endomorphism of F $(\mathrm{char}(F) = p)$ is, $\sigma_F : F \to F \quad x \mapsto x^p$

2) Say F is perfect if σ_F is an isomorphism

eg. 1) $\mathbb{F}_p$. ; 2). $\mathbb{F}_p\left(t, t^{\frac{1}{p}}, t^{\frac{1}{p^n}}, n \in \mathbb{N}\right)$. ; 3) $\mathbb{F}_p(t)$ is imperfect

Prop. Algebraic extensions of perfect fields are still perfect

pf: It suffices to prove: K/F (algebraic), $\forall \alpha \in K$. admits a pth root in $E := F(\alpha)$.

We have the following field extensions:

$$\begin{array}{ccc} & E & \\ \sigma(E) & | & \\ | & F & \\ \sigma(F) & & \end{array}$$

$\Rightarrow [E : \sigma(E)] \cdot [\sigma(E) : \sigma(F)] = [E : F][F : \sigma(F)]$

we have $[\sigma(E) : \sigma(F)] = [E : F]$ (σ is isomorphisms on vector-spaces).

then $[E : \sigma(E)] = [F : \sigma(F)] = 1$.

$\Rightarrow E = \sigma(E)$ ▨

Remark. For a field F of char $p > 0$. We define its perfection $F^{perf} := \bigcup_{n \geq 0} \sigma^{-n}(F)$.

tensions and finite fields.

~~K/E is separable if~~

If F is a field, $f(x) \in F[x]$ then f has no repeated roots iff $(f(x), Df(x)) = (1)$

pf: $\Leftarrow$ if $(f(x), D(f(x))) = (1)$ then $\exists\ p(x), q(x) \in F[x]$

s.t. $p(x)f(x) + q(x)D(f(x)) = 1$. If $(x-a)^2 \mid f(x) \Rightarrow (x-a) \mid D(f)(x)$

$\Rightarrow x-a \mid 1$. contradict!

$\Rightarrow$ if $f(x) = (x-a_1)\cdots(x-a_n)$ $a_i \neq a_j \in K$ $(i \neq j)$

$$D(f)(x) = \sum_{i=1}^{n} \prod_{j \neq i} (x-a_j) \Rightarrow D(f)(a_j) \neq 0 \quad \forall\ (\leq) \leq n$$

thus $\gcd(f(x), D(f)(x)) = 1 \Rightarrow (f(x), D(f)(x)) = (1)$. □

Definition: $f(x)$ has only simple roots in its splitting field, we say f is separable. otherwise it's called inseperable

($f(x)$ is irreducible in $F[x]$ and)

Then we have the corollary:

1) If $\mathrm{char}(F) = 0$. then all irreducible polynomials are seperable

2) If $\mathrm{char}(F) = p$. then if $f(x)$ is inseparable, then $f(x) = g(x^p)$ for some $g \in F[x]$ irreducible

Moreover, F is imperfect.

pf: 1) is trivial; 2) $D(f)(x) = a_1 + 2a_2 x + \cdots + n a_n x^{n-1} = 0 \Rightarrow i a_i = 0 \Rightarrow a_i = 0$ $p \nmid i$

$\Rightarrow f(x) = a_0 + a_p x^p + a_{2p} x^{2p} + \cdots = g(x^p)$ $g(x) = a_0 + a_p x + a_{2p} x^2 + \cdots$ is irreducible

And we have $f(x) = b_0^p + b_1^p x^p + \cdots = (b_0 + b_1 x + \cdots)^p$, when F is perfect ▨

~~Remark: That means~~

Definition: Let K be an algebraic extension of F

(1) $\alpha \in K$ is seperable iff $m_{\alpha, F}(x)$ is seperable.

(2) K is seperable over F iff $\forall\ \alpha \in K$., α is seperable.

Thm: (1) If α is seperable over F, then $F(\alpha)$ is a seperable extension over F.

(2) If K/E, E/F are seperable extensions. then K/F is seperable.

pf: (skipped).

Thm: A finite seperable extension is generated by one element.

Next, we'll give the conception of Galois group and the proof of the Gal... theory

Let G be the $|Gal(\quad)|$ thm

Definition: 1) Suppose K/F, $Gal(K/F) = \{\sigma : K \to K \mid \sigma \text{ is an isomorphism and } \sigma|_F = id\}$ called the Galois group of K/F

2) K/F finite, then K/F is called Galois extension iff K/F is normal and seperable

Thm (Finite Galois correspondance): Let K/F be a finite Galois extension, note that $G = Gal(K/F)$

let $\mathcal{M} = \{M \mid F \subseteq M \subseteq K\}$, $S = \{H \mid H \leq G\}$, then there exists an one to one correspondance,

let $\Phi : \mathcal{M} \to S \quad M \mapsto Gal(K/M)$.

$\Psi : S \to \mathcal{M} \quad H \mapsto K^H = \{x \in K \mid h(x) = x, \forall h \in H\}$

then $\Phi \cdot \Psi = id_S$, $\Psi \circ \Phi = id_{\mathcal{M}}$

$K \longleftrightarrow \{1\}$

$M \longleftrightarrow Gal(K/M)$

$F \longleftrightarrow Gal(K/F)$

Moreover, ① $[K^{\bullet} : K^H] = |H|$; $[K^H : F] = [G : H]$

② M/F is Galois iff $Gal(K/M) \trianglelefteq Gal(K/F)$

③ If $Gal(K/M) \trianglelefteq Gal(K/F)$, then $Gal(M/F) \cong \dfrac{Gal(K/F)}{Gal(K/M)}$

pf: ① $\Psi \circ \Phi = id_M \Leftrightarrow K^{Gal(K/M)} = M$.

One one hand, $M \subseteq K^{Gal(K/M)}$ is obvious

On the other hand, suppose that $M = K^{Gal(K/M)} \neq M$

choose $\alpha \in M' \setminus M$. then $\deg Irr_{M,\alpha}(x) \geq 2$

Since K/M is normal and seperable, $\Rightarrow \exists \beta \in K$. s.t. $\beta \neq \alpha$. $Irr_{M,\alpha}(\beta) = 0$

Hence there exists a M-isomorphism $\varphi : M(\alpha) \longrightarrow M(\beta)$, $\alpha \mapsto \beta$.

Extend the φ to the bigger field K. $\overline{\varphi} : K \to K$

We have

$$\begin{array}{ccc} K & \xrightarrow{\overline{\varphi}} & K \\ | & & | \\ M(\alpha) & \xrightarrow{\varphi} & M(\beta) \\ | & & | \\ M & = & M \end{array}$$

$\Rightarrow \overline{\varphi} \in Gal(K/M)$

thus $\overline{\varphi}(\alpha) = \alpha \neq \beta$ contradiction

② $\Phi \circ \Psi = id_S \Leftrightarrow H = Gal(K/K^H)$

On one hand: $H \leq Gal(K/K^H)$ is obvious

On the other hand: suppose $K = K^H(\alpha)$ (by the last thm and K/K^H is Galois)

Consider $Q(x) = \prod_{h \in H} (x - h(\alpha)) \in K^H[x]$ and $Q(\alpha) = 0$

let $P(x) = Irr_{K^H,\alpha}(x)$ then $P(x) \mid Q(x)$

$\Rightarrow [K : K^H] = \deg P \leq \deg Q \leq |H|$

One It's easy to figure out, for all finite extension L/E, $|Gal(L/E)| \leq [L : E]$

~~suppose $L = E(\alpha_1, \dots, \alpha_n)$. then $|Gal(L/E)|$ $\begin{cases} \\ \end{cases}$ $[L:E] = \prod_{i=1}^{n} \deg I_{r_i, \alpha_i, E}(x)$~~

Ω be the algebraic closure of E

then $|Gal(L/E)| \le |\{L \text{ embedding } \to \Omega\}| \le [L:E]$

thus $|H| \le |Gal(K/K^H)| \le [K:K^H] \le |H| \Rightarrow |H| = |Gal(K/K^H)| = [K:K^H]$ □

Some specific caculation thm.

Prop. 1. $f(X) \in F[X]$ is seperable, K_f is the splitting field. Ω is the root ret $\{\alpha_1, \dots, \alpha_n\}$.
then $f(X)$ is irreducible in $F[X]$ iff $G_f = Gal(K_f/F)$-action on Ω is transtive.

2. Define the discriminant of $f(x) \in F[X]$ irreducible. $\Delta(f) = a^{2n-2} \prod_{1 \le i < j \le n} (\alpha_i - \alpha_j)^2$.
where $f(x) = a \prod_{i=1}^{n} (x - \alpha_i)$, $\text{char}(F) \neq 2$.
then $G_f \le A_n$ iff $\Delta(f) \in (F^\times)^2$.

3. let $q = p^r$. p is a prime. then $\mathbb{F}_{q^n}/\mathbb{F}_q$ is a Galois extension and
$Gal(\mathbb{F}_{q^n}/\mathbb{F}_q) = \langle \varphi_q \rangle \cong \mathbb{Z}/n\mathbb{Z}$. where $\varphi_q : \mathbb{F}_{q^n} \to \mathbb{F}_{q^n}$. is called q-Frobenius automorphism
$x \mapsto x^q$

Corallory 1. $m|n$ then $\mathbb{F}_{q^n}/\mathbb{F}_{q^m}/\mathbb{F}_q$.

2. $f(X) \in \mathbb{F}_q[X]$ irreducible. $\deg f = d$. If $f(\alpha) = 0$. 则 $\mathbb{F}_q(\alpha) = \mathbb{F}_{q^d}$.
All roots of f are $\alpha, \alpha^q, \alpha^{q^2}, \dots, \alpha^{q^{d-1}}$

3. Suppose $f(x) \in \mathbb{F}_q[X] = f_1(x) \cdots f_r(x) \in \mathbb{F}_q[x]$ $\deg f_i = d_i$.
then ~~$\forall$~~ the splitting field of f on $\mathbb{F}_q$ is $\mathbb{F}_{q^m}$. where $m = \text{lcm}(d_1, \dots, d_r)$

talk several properties on rings and ideals.

Let $I_1, \dots, I_k$ be ideals of a communtative ring R.

Then the natural map

$$\phi: R \longrightarrow R/I_1 \times \dots \times R/I_k$$
$$x \longmapsto (x \bmod I_1, \dots, x \bmod I_k).$$ is a ring homomorphism with kernel $I_1 \cap \dots \cap I_k$

If $I_1, \dots, I_k$ are pairwise comaximal (that is: $I_i + I_j = R$).

then (1) ϕ is surjective.

(2) $I_1 \cap \dots \cap I_k = I_1 I_2 \dots I_k$. and thus $\dfrac{R}{I_1 I_2 \dots I_k} \cong \dfrac{R}{I_1 \cap \dots \cap I_k} \cong \dfrac{R}{I_1} \times \dots \times \dfrac{R}{I_k}$

pf: First, $\phi(x+y) = \phi(x) + \phi(y)$, $\phi(xy) = \phi(x)\phi(y)$.

$\ker \phi = \{x \mid \phi(x) = 0\} = \{x \mid x \in \bigcap_{i=1}^{k} I_k\}$.

Next, consider the situation when $k = 2$.

$I_1 I_2 \subseteq I_2$. $I_1 I_2 \subseteq I_1 \Rightarrow I_1 I_2 \subseteq I_1 \cap I_2$

Note that $R = I_1 + I_2$. $1 = a_1 + a_2$. $a_i \in I_i$ for $b \in I_1 \cap I_2$.

$b = ba_1 + ba_2 \in I_1 I_2$. $\Rightarrow I_1 I_2 = I_1 \cap I_2$ ..... (1).

Then, $\phi(a_1) = (0, 1)$. $\phi(a_2) = (1, 0)$. thus

$\phi(a_1 x_2 + a_2 x_1) = (x_1 \bmod I_1, x_2 \bmod I_2)$. $\Rightarrow \phi$ is surjective.

In general, $\phi: R \to R/I_1 \times R/I_2 \dots I_k \twoheadrightarrow R/I_1 \times \dots \times R/I_k$.

where $I_2 \dots I_k = I_2 \cap \dots \cap I_k$

then we suffice to check $I_1 + I_2 \dots I_k = R$

$1 = a_i + b_i$ $a_i \in I_1$, $b_i \in I_i$

then $1 = (a_2 + b_2) \dots (a_k + b_k) = a_2 \dots a_k +$ product with some $a_i +$ $\underbrace{b_1 \dots b_k}_{\text{in } I_2 \dots I_k}$ $\square$.

p.s. That is Chinese remainder theorem in commutative rings.

$\eta: F \xrightarrow{\sim} F'$ be an isomorphism. $f(x) \in F[x]$. Put $f'(x) := \eta(f(x)) \in F'[x]$.

E is a splitting field of $f(x)$ over F. E' is a splitting field of $f'(x)$ over F'.

Then there exists an isomorphism $\sigma: E \xrightarrow{\sim} E'$ restricting to $\eta: F \to F'$.

Proof: (i). We construct a splitting field K of $f(x)$ over F.

$\deg f = 1$. $K = F$

Suppose $\deg f \leq n-1$ we have the splitting fields

$\deg f = n$, let $p(x)$ be an irreducible factor of $f(x)$.

Then $L := F[x]/(p(x))$ is a field of degree $\deg p(x) \leq n$,

over which $p(x)$ has a zero θ.

Thus $f(x) = (x-\theta)g(x)$ over L. ♯

by inductive hypothesis, $g(x)$ over L admits a splitting field K

note: $K/L/F$, then K is also a splitting field of $f(x)$ over F.

(ii) We take K in (i) as an explicitly constructed splitting field

We'll prove the following commutative diagram: $E' \supseteq F'$ over which $\eta(f(x))$ splits completely.

$$\begin{array}{ccccc} E & \xleftarrow{\sigma_1} & K & \xrightarrow{\sigma'} & E' \\ \cup & & \cup & & \cup \\ F & = & F & \xrightarrow[\cong]{\eta} & F' \end{array}$$

We'll just ~~prove~~ construct the right diagram:

(iii). Make an induction on $\deg(f)$:

$\deg f = 1$. $K = F$. $E' = F'$.

Suppose $\deg f \leq n-1$, we have constructed such σ'.

then $L = F[x]/(p(x)) \xrightarrow{?} E'$ As $\eta(p(x)) = p'(x)$ has a zero in E', say $\alpha \in E'$.

$F \xrightarrow{\eta} E'$

define $\sigma_L': L = F[x]/(p(x)) \longrightarrow E'$

$\theta = x + (p(x)) \longmapsto \alpha$

$F \xrightarrow{\eta} E'$

~~where $f(x) = g(x)(x-\theta)$ ... over L.~~ ~~$g(x)$ admits a splitting field K~~

~~and $\sigma': K \to E' \xrightarrow{\sim} E'$~~

~~then σ_L' ... $\sigma_L'(L)$~~

let $f(x) = (x-\theta)g(x)$ over L

then

$$\begin{array}{ccc} K & \xrightarrow{\sigma'} & E' \\ \cup & & \cup \\ L & \xrightarrow[\cong]{\sigma_L'} & \sigma_L'(L) \end{array}$$

K is a splitting field ~~over~~ of $g(x)$ over L.

by induction, $\sigma': K \longrightarrow E'$ is a homomorphism

□

, a_n be ideals & let p be a prime ideals. containing $\bigcap_{i=1}^{n} a_i$.

$p \supseteq a_i$ for some a_i If $p = \bigcap a_i$. then $p = a_i$ for some i.

2) Let $p_1, \dots, p_m$ be prime ideals. and let a be an ideal contained in $\bigcup_{i=1}^{n} p_i$.

Then $a \subseteq p_i$ for some a_i.

pf: (1). Suppose not. take elements $x_i \in a_i \setminus p$. for each i.

Since p is a prime ideal,

$x_1 x_2 \dots x_n \notin p$. but $x_1 \dots x_n \in \bigcap_{i=1}^{n} a_i \subseteq p$. contradiction!

If moreover $p = \bigcap a_i \supseteq a_j$ for some j then $p = a_j$ for some j.

(2). We prove by induction on n that

$$a \nsubseteq p_i \quad i=1,\dots,n \implies a \nsubseteq \bigcup_{i=1}^{n} p_i$$

$n=1$ is clear.

Suppose that we have proved $n-1$.

By inductive hypothesis. we may find $x_i \in a$. such that

$x_i \notin p_1 \cup \dots \cup p_{i-1} \cup p_{i+1} \cup \dots \cup p_n$.

if for some x_i, $x_i \in p_i$. then $x_i \notin \bigcup_{i=1}^{n} p_i$ we get the conclusion.

if $x_i \in p_i$ for each i.

Then we consider $y = \sum_{i=1}^{n} x_1 \dots x_{i-1} x_{i+1} \dots x_n$

clearly $y \in a$. and $y \notin p_i$ for each i.

In fact, the sum except $x_1, \dots x_{i-1} x_{i+1} \dots x_n$ belongs to p_i.

but p_i is a prime ideal. $\Rightarrow$ $x_1 \dots x_{i-1} x_{i+1} \dots x_n \notin p_i$.

$\Rightarrow$ This completes the proof. $\square$.